

MAGMA: inference and prediction using multi-task Gaussian processes with common mean

Ben Kabongo ben.kabongo_buzangu@ens-paris-saclay.fr

Hugo Queniat hugo.queniat@telecom-paris.fr

Paul Castéras paul.casteras@student-cs.fr

We worked on the article **MAGMA: inference and prediction using multi-task Gaussian processes with common mean** written by Arthur Leroy, Pierre Latouche, Benjamin Guedj, and Servane Gey [1].

The model proposed in this article is a multi-task Gaussian process, applied to time series forecasting, where processes do not share a common covariance matrix and zero mean, as in most previous work, but share a common mean. The model is trained with the EM algorithm to calculate its parameters.

The model

$$\mathbf{y}_i(t) = \mu_0(t) + f_i(t) + \varepsilon_i(t) \quad (1)$$

with :

$$\mu_0(\cdot) \sim \mathcal{GP}(m_0(\cdot), k_{\theta_0}(\cdot, \cdot)), f_i(\cdot) \sim \mathcal{GP}(0, c_{\theta_i}(\cdot, \cdot)). \quad (2)$$

- E step : The hyper-posterior distribution of μ_0 remains Gaussian: $p(\mu_0 | \{\mathbf{y}_i\}_i, \hat{\Theta}) = \mathcal{N}(\mu_0(\mathbf{t}); m_0(\mathbf{t}), \hat{\mathbf{K}}^t)$

- M step : Equivalent to solving the following problems

$$\hat{\theta}_0 = \arg \max_{\theta_0} \mathcal{L}^t(\hat{m}_0(\mathbf{t}); m_0(\mathbf{t}), \mathbf{K}_{\hat{\theta}_0}^t)$$

$$\forall i, (\hat{\theta}_i, \hat{\sigma}_i^2) = \arg \max_{\theta_i, \sigma_i^2} \mathcal{L}^{t_i}(\mathbf{y}_i; \hat{m}_0(\mathbf{t}), \Psi_{\theta_i, \sigma_i^2}^{t_i})$$

where $\mathcal{L}^t(\mathbf{x}; \mathbf{m}, \mathbf{S}) = \log \mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{S}) - \frac{1}{2} \text{Tr}(\hat{\mathbf{K}}^t \mathbf{S}^{-1})$.

Prediction

For a new individual $*$, $\{\mathbf{t}_*, y_*(\mathbf{t}_*)\}$, we want to predict its evolution at future timestamps $\mathbf{t}^p$.

- Compute the posterior inference on the mean process $\mu_0(\cdot)$ whose distribution remains Gaussian :

$$p(\mu_0(\mathbf{t}_*^p) | \{\mathbf{y}_i\}_i) = \mathcal{N}(\mu_0(\mathbf{t}_*^p); \hat{m}_0(\mathbf{t}_*^p), \hat{\mathbf{K}}_*^p)$$

- We then observe that, for individual $*$ with $\mathbf{t}_*^p = \mathbf{t}_* \cup \mathbf{t}^p$:

$$p(y_*(\mathbf{t}_*^p) | \{\mathbf{y}_i\}_i) = \mathcal{N}(y_*(\mathbf{t}_*^p); \hat{m}_0(\mathbf{t}_*^p), \hat{\mathbf{\Gamma}}_*^p)$$

- When dealing with different hyper-parameters in the model, we maximize this likelihood to obtain the optimal $\hat{\Theta}_*$.

- Finally, the posterior distribution becomes

$$p(y_*(\mathbf{t}^p) | y_*(\mathbf{t}_*), \{\mathbf{y}_i\}_i) = \mathcal{N}(y_*(\mathbf{t}^p); \hat{\mu}_0^p, \hat{\mathbf{\Gamma}}^p)$$

where mean vectors and covariance matrices are now known.

Experiments

		Estimation of μ_0	Prediction
Common HP	Common grid	65.867	418.062
	Uncommon grid	67.789	132.734
Different HP	Common grid	38.557	63.293
	Uncommon grid	149.251	57.474

Table: Loss MSE of μ_0 mean estimation by MAGMA and loss MSE of prediction on synthetic data, for the different possible model configurations.

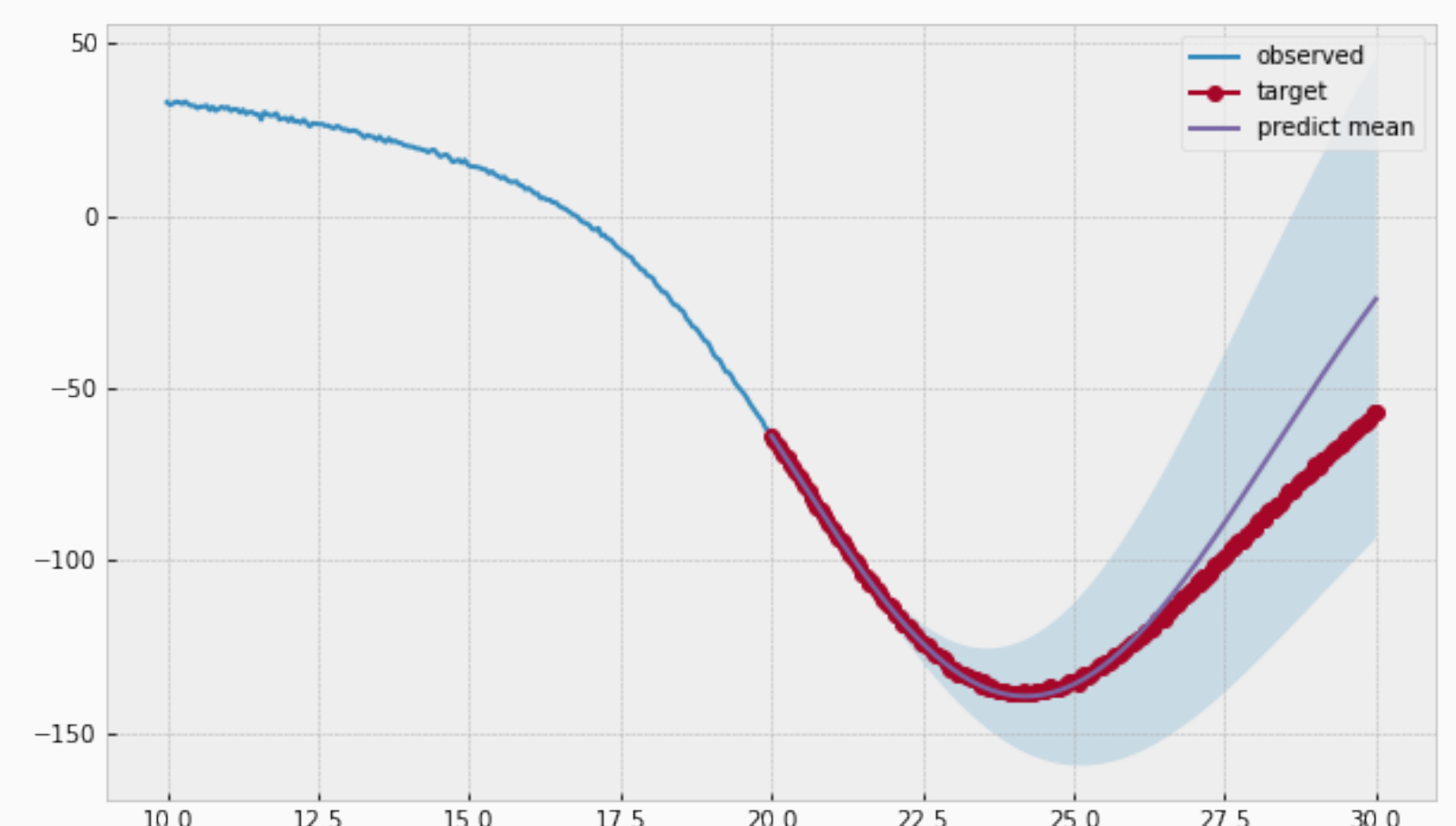
		MAGMA	GP
ECG5000	M = 10 Common HP	0.366 (0.020)	0.376 (1.276×10^{-15})
	Different HP	0.360 (0.053)	0.376 (1.276×10^{-15})
	M = 20 Common HP	0.361 (0.016)	0.374 (1.110×10^{-16})
	Different HP	0.359 (0.064)	0.374 (1.110×10^{-16})
Traffic	M = 5 Common HP	6.005×10^{-3} (2.768×10^{-4})	6.009×10^{-3} (3.642×10^{-17})
	Different HP	5.988×10^{-3} (1.775×10^{-4})	6.009×10^{-3} (3.642×10^{-17})
	M = 10 Common HP	5.995×10^{-3} (2.152×10^{-4})	6.021×10^{-3} (3.729×10^{-17})
	Different HP	5.962×10^{-3} (1.574×10^{-4})	6.021×10^{-3} (3.729×10^{-17})

Table: Loss MSE on real datasets. The models are trained on 60% of each individual, and predictions are made and evaluated on the remaining 40%. **The mean of individuals for prediction is given by the dataset mean.** The mean MSE value and standard deviation are given in brackets.

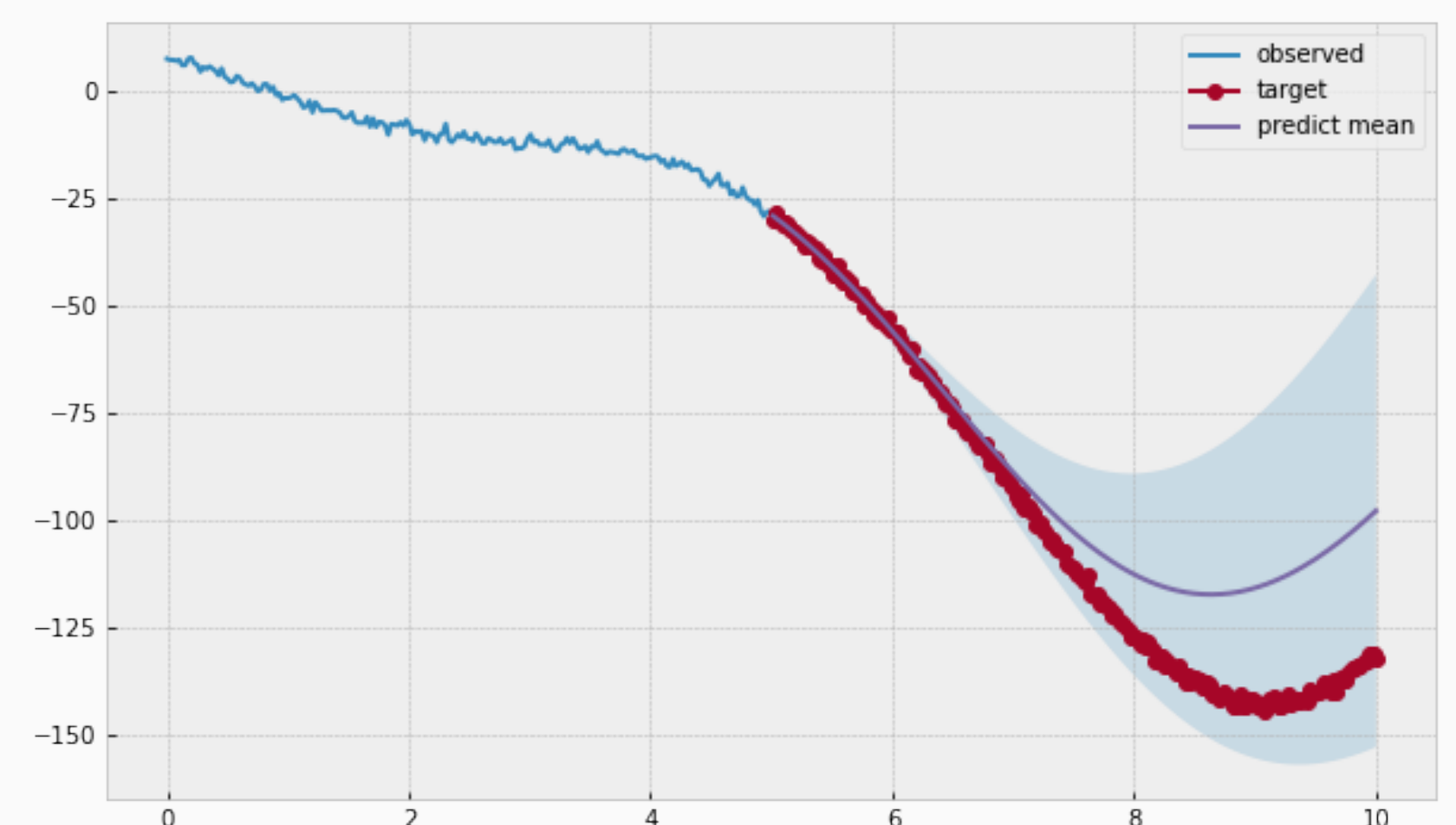
Discussion

- MAGMA 's MSE is always lower than that of GPs.
- The model relies on a common Mean Process : it showcases a higher variance in the estimation of the mean compared to GP.
- MAGMA showed a relatively low improvement over all our experiments but proved more demanding in terms of computing resources.

Both types of prediction



A prediction of type I



A prediction of type II

References

- [1] Arthur Leroy et al. "MAGMA: inference and prediction using multi-task Gaussian processes with common mean". In: *Machine Learning* 111 (5 2022).